



Genomic characteristics, virulence, and antimicrobial resistance in avian pathogenic *Escherichia coli* MTR_BAU02 strain isolated from layer farm in Bangladesh

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ARTICLE INFO

Article history:

Received 6 April 2022

Revised 29 May 2022

Accepted 1 June 2022

Available online 6 June 2022

Editor: Stefania Stefani

Keywords:

Poultry

Escherichia coli

Pathogenic

Antibiotic Resistance

Virulence

ABSTRACT

Background: Colibacillosis, caused by avian pathogenic *Escherichia coli* (APEC), is one of the most significant infectious diseases affecting poultry worldwide.

Objectives: This study aimed to determine the genomic diversity, virulence factor genes (VFGs), and antimicrobial resistance genes (ARGs) in the APEC MTR_BAU02 strain isolated from a layer chicken using whole-genome sequencing (WGS).

Methods: Paired-end (2 × 250) WGS was performed using Illumina MiSeq sequencer (Illumina, San Diego, CA) and de novo assembly was performed using SPAdes. Core genome multilocus sequence typing (cgMLST) analysis between APEC MTR_BAU02 and all of the ST1196 *E. coli* strains retrieved from the National Center for Biotechnology Information (NCBI) GenBank database was performed using the BacWGSTdb 2.0 server. We utilized different databases to detect ARGs, VFGs, and genomic functional features of the APEC MTR_BAU02 strain.

Results: The complete genome of APEC MTR_BAU02 consists of 94 contigs comprising 4,924,680 bp (51.1% guanine-cytosine [GC] content), including 4681 protein-coding sequences, one chromosome, and one plasmid, and was assigned to ST1196. The closest relatives of APEC MTR_BAU02 were four isolates originating from human clinical specimens (diarrhetic stool) in Bangladesh and two clinical isolates originating from chicken in India, which differed by 694 core genome multilocus sequence typing (cgMLST) alleles. One hundred and twenty-two ARGs and 92 VFGs were identified in the APEC MTR_BAU02 genome. Metabolic functional annotations detected 380 SEED subsystems including genes coding for carbohydrate metabolism, protein metabolism, cofactors, vitamins, prosthetic groups and pigments, respiration, membrane transport, stress response, motility and chemotaxis, and virulence, disease, and defense.

Conclusion: This study reports the genome sequence of a multidrug-resistant APEC strain isolated from layer birds in Bangladesh. The ARGs and VFGs, widespread in APEC MTR_BAU02, are similar to those found in human isolates, and highlight the growing threat of antimicrobial resistance in both poultry and humans.

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1. Introduction

Escherichia coli is a Gram-negative, dangerous, opportunistic pathogen in the poultry industry [1]. This ubiquitous pathogen has adaptive ability in diverse ecological niches and can induce enteric and extraintestinal infections in animals and humans [1,2]. Moreover, strains of *E. coli* are thought to be virulent in humans and

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animals, including commercial poultry, and hold zoonotic potential [2,3]. Colibacillosis, caused by avian pathogenic *E. coli* (APEC), is the most common bacterial disease of chickens reared for egg and meat production [1,4], and poses a profound threat to one of humankind's cheapest sources of high-quality animal protein [5]. This is one of the most significant infectious diseases affecting the global poultry industry [1,4]. APEC exists as a commensal component of the avian gut microbiota but emerges to take many forms, with a variety of systemic infections [1,5]. Even though different risk factors have been identified, including chicken immunological immaturity and stress for colibacillosis, the disease has proved difficult to control. Strains of *E. coli* also have zoonotic significance since they are known to cause infections in both humans and animals, including birds [1,6]. Currently, one of the greatest challenges for APEC is the difficulty of understanding their pathogenicity [4]. In addition, their pathogenicity appears to be linked to a heterogeneous mix of plasmid and chromosomal genes involved in bacterial adhesion, invasion, toxicity, antibiotic resistance, survival and metabolism under stress [7,8], and the emergence of virulent APEC from a background of commensal gut-dwelling *E. coli* [8,9].

Antimicrobial resistance (AMR) is an ever-increasing public health crisis [10]. Multidrug-resistant (MDR) APEC strains can contaminate the food supply from farm to fork through eggs, meat, and other contaminants, and thus pose a severe threat to the consumer's health [1,11]. The indiscriminate use of antibiotics in poultry production may have contributed to drug resistance in APEC. There are reports that resistant *E. coli* strains might enhance AMR in other organisms (pathogenic and non-pathogenic) within the gastrointestinal tract of the chicken [1,12], and help in the transmission and dissemination of MDR strains and ARGs between animal and human pathogens [1–3]. Earlier studies have explored the association between pathogenic traits of APEC and their repertoire of virulence factor genes (VFGs) [1,8]; however, potential pathogenic traits associated with APEC strains warrant further research into the potential roles of VFGs, MDR properties, and their crosstalk with specific hosts. Technological advent in high-throughput whole-genome sequencing has dramatically revolutionized clinical microbiology and the investigation of the population genomics of pathogen evolution [8]. However, understanding the spread of APEC in poultry populations remains challenging because of uncertainty regarding the transmission potential of a few globally distributed epidemic clones vs. a diverse assemblage of disease-causing lineages [7]. Genes contributing to APEC virulence are less well-described than those in human pathotypes [8].

In order to investigate the role of APEC in the pathophysiology of laying birds, as well as its contributions in spreading ARGs and VFGs, the reported APEC MTR_BAU02 strain was isolated from laying birds belonging to different farms in Bangladesh. Here, we report the whole genome of the APEC MTR_BAU02 strain and its functional genomics to elucidate the mechanisms through which this pathogen shows its resistant and virulent phenotypes, and contributes to the spread of ARGs and VFGs and pathogenesis in layer farms in Bangladesh.

2. Materials and methods

The APEC MTR_BAU02 strain was recovered from faecal sample of a sick laying bird with symptoms of colibacillosis in November of 2019. The sample was inoculated and incubated overnight on eosin methylene blue (EMB) agar plates at 37°C. The colonies showing characteristic metallic sheens on the EMB agar plates (three to five colonies from each sample) were considered indicative of *E. coli*. These colonies were then subjected to morphological study by Gram staining, biochemical tests (indole, methyl-red, catalase, citrate, and Voges-Proskauer) for confirmatory identification of *E. coli*. The final confirmation of the isolation of *E.*

coli was performed by polymerase chain reactions (PCRs) targeting the *E. coli* 16S rRNA gene [11]. APEC MTR_BAU02 strain was subjected to antimicrobial susceptibility testing by the disc diffusion method for ten commonly used antibiotics of different classes: penicillins (ampicillin), amphenicols (chloramphenicol), fluoroquinolones (ciprofloxacin, enrofloxacin, and norfloxacin), polypeptides (colistin), macrolides (erythromycin), aminoglycosides (gentamicin and streptomycin), and tetracycline (tetracycline).

The genomic DNA from the APEC MTR_BAU02 strain was extracted using a commercial DNA extraction kit, QIAamp DNA Mini Kit (QIAGEN, Hilden, Germany), in accordance with the manufacturer's instructions. Concentration and purity of DNA were measured using NanoDrop 2000 UV-Vis Spectrophotometer (Thermo Fisher, Waltham, MA). DNA extracts with A260/280 and A260/230 ratios of ~1.80 and 2.00 to 2.20, respectively, were considered as high-purity DNA samples. Finally, harvested DNA were visualized on 1% (w/v) agarose gel and chosen for sequencing based on their high purity and adequate concentration. DNA libraries were prepared using the Nextera DNA Flex Library Prep Kit (Illumina, San Diego, CA) according to the manufacturer's protocol. The whole-genome sequencing (WGS) of the prepared libraries was performed through Illumina MiSeq sequencer (Illumina, San Diego, CA) using a 2 × 250 paired-end protocol.

Trimmomatic [13] was used to check the quality FASTQ reads and trim low-quality residues [4]. Trimmed reads were de novo assembled using SPAdes 3.15.2 [14]. Orphaned reads resulting from trimming (i.e., reads where only one read of the pair survived) were provided to the assembler as single-end reads. Trimmed paired-end and orphaned reads were mapped against the de novo assembly, the Sakai *E. coli* O157:H7 reference genome (NCBI: National Center for Biotechnology Information, accession NC_002695.2) [15], using Bowtie2 2.3.0 [16] with the '–sensitive' and '–end-to-end' settings. The genome sequence was automatically annotated by the NCBI Prokaryotic Genomes Annotation Pipeline (PGAP). The APEC MTR_BAU02 genome was compared with the genomes of reference strains APEC01 (GenBank accession number NC_008563) [17] and APEC078 (GenBank accession number CP004009) [5]. The circular visualization of the APEC MTR_BAU02 genome was performed using CGViewer [18]. In silico MLST analysis and bacterial source tracking using a cgMLST strategy were performed using BacWGSTdb 2.0 server [19]. The Comprehensive Antibiotic Resistance Database (CARD) [20] and VirulenceFinder [21,22] databases were used to detect ARGs and VFGs, respectively. The CARD (<https://card.mcmaster.ca>) integrated within AMR++ pipeline identified the respective genes or protein families coding for ARG markers in the APEC MTR_BAU02 strain [20]. We utilized SnapGene Viewer web tool (<https://www.snapgene.com/snapgene-viewer/>) to visualize the virulence plasmid of APEC MTR_BAU02. The draft genome of APEC MTR_BAU02 was also annotated using RAST (Rapid Annotation using Subsystem Technology) server [23] to detect the genomic functional features.

3. Results and discussion

The de novo genome assembly revealed that the APEC MTR_BAU02 genome was 4,924,680 bp in size with 4,497,218 bp DNA coding sequences, and a coverage of 38.98x. The draft genome had 2,516,511 bp DNA sequences assigning for guanine + cytosine (G + C) content of 51.1% (Table 1). The complete genome of APEC MTR_BAU02 possessed 94 contigs, including 332,120 bp contig N50, 6 contig L50, and one plasmid. Of the detected contigs, the largest and smallest contig size were 4,49,968 bp and 234 bp, respectively. Notably, preliminary sequence analysis revealed 41 insertion sequences (ISs), 14 predicated genomic islands (GIs), seven prophage-related sequences, two CRISPR arrays, and one plasmid (Table 1). The genome of APEC MTR_BAU02 encodes

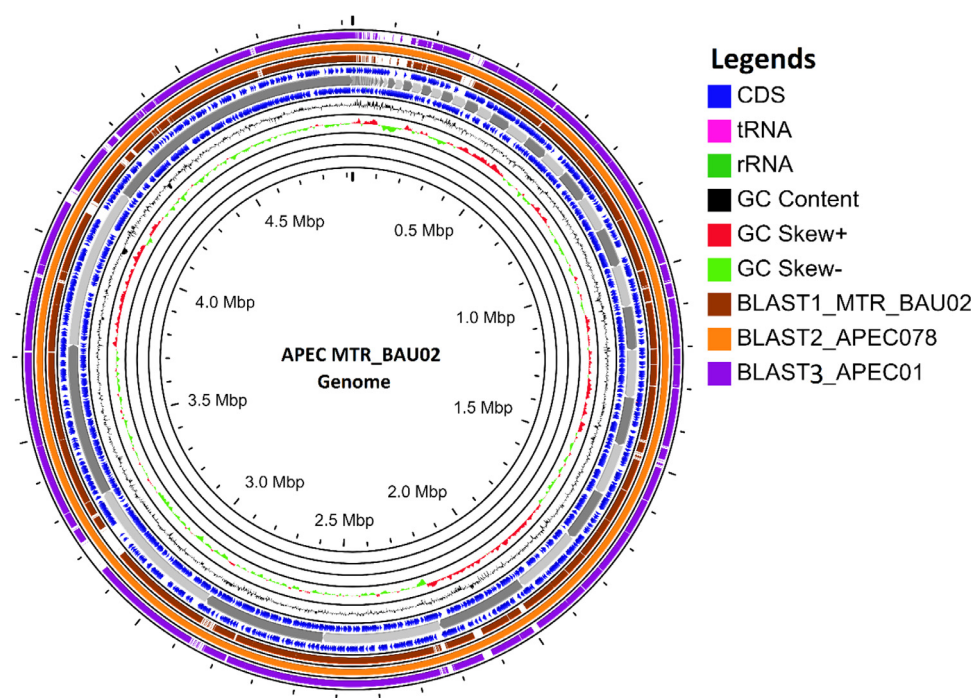


Fig. 1. Circular genome representation of APEC MTR_BAU02 compared with APEC01 (NC_008563) and APEC078 (CP004009). The outermost ring: *Escherichia coli* BLAST 3 results for APEC01 (purple), followed by *E. coli* BLAST 2 results for APEC078 (solid yellow) and *E. coli* BLAST 3 results for MTR_BAU02 (purple) representing the positions covered by the BLAST comparative alignment results (BLASTN), G + C content (black), G + C positive skew (red), and G + C negative skew (green). Image created using CGview Server.

Table 1

Avian pathogenic *Escherichia coli* (APEC) strain MTR_BAU02 genome characteristics.

Attribute(s)	Value	% of Total ^a
Genome size (bp)	4,924,680	100.00
DNA coding (bp)	4,497,218	91.32
Genome coverage	38.98x	38.98
DNA G + C (bp)	2,516,511	51.1
Total contigs	94	
Contig N50 (bp)	332,120	6.74
L50	6	
Total genes	4,803	100.00
Coding sequences	4,681	97.46
Protein coding genes	4,571	95.17
RNA genes	120	2.5
tRNA genes	81	1.69
rRNA genes	28	0.58
Pseudo genes	112	2.33
Genes with function prediction	3,940	82.03
Genes assigned to SEED subsystems	3,512	73.12
Number of subsystems	380	
Genes with Pfam domains	4,214	87.74
Genes with signal peptides	386	8.04
Genes with transmembrane helices	1,084	22.57
CRISPR arrays	2	
Number of plasmids	1	

bp, base pair.

^a Total based on either the size of the genome in bps or the total number of genes in the annotated genome.

for 4801 genes, including 4681 protein coding sequences (CDSs), 28 rRNAs (5S = 8, 16S = 5, 23S = 15), 81 tRNAs, 11 ncRNAs, and 88 pseudogenes (Fig. 1).

Antimicrobial susceptibility testing showed that APEC MTR_BAU02 isolates were resistant to ampicillin and tetracycline (100%), followed by chloramphenicol and erythromycin (97.2%), enrofloxacin (55.5%), norfloxacin and ciprofloxacin (50.0%), streptomycin (19.4%), colistin (11.1%), and gentamicin (8.3%). The

resistome of the APEC MTR_BAU02 genome consisted of genes conferring resistance to aminoglycosides (*aadA1*), β -lactams (*blaTEM-106*), phenicols (*cmlA1*), macrolides [*mdf(A)*], quinolones (*qnrS1*), sulfamethoxazole (*sul3*), and tetracyclines [*tet(A)*]. This resistome repertoire was related to several important genomic functions of APEC MTR_BAU02 deliberating resistance to multiple antibiotics. For instance, genes coding for multidrug efflux pump system (*MdtABC-TolC*, 7.77%; *EmrAB-OMF*, 3.88%; *EmrKY-TolC*, 3.88%; *AcrEF-TolC*, 2.91%; *MdtC*, 2.91%; *EmrB*, 2.91%; and *MdtB*, 2.91%); acid resistance, osmotic adaptation, and drug resistance (*EvgA*, 3.88%); and aminoglycosides efflux system (*AcrD*, 2.91% and *AcrAD-TolC*, 2.91%) were revealed. Efflux pump conferring resistance to multiple antibiotics was found as the predominating resistance class in the genome of APEC MTR_BAU02.

The APEC MTR_BAU02 strain belongs to ST1196 according to Achtman's MLST scheme for *E. coli*. To reveal the genomic epidemiological characteristics of *E. coli* strains in a global context, the phylogenetic relationships between APEC MTR_BAU02 and a total of 39 other ST1196 *E. coli* strains currently deposited in the NCBI GenBank database were analysed (Fig. 2A). cgMLST analysis revealed that the closest relatives of APEC MTR_BAU02 were four ST1196 strains originating from diarrhetic human stool in Bangladesh (accession numbers APXI01, APZN01, AQDV01, and AQEY01) and two clinical isolates originating from chicken in Meghalaya and Mysore, India (accession numbers WCFA01 and RDCD01), which differed by 694 cgMLST alleles (Fig. 2A). These Indian strains also harboured different ARGs, including *aadA1*, *aadA2*, *blaCTX-M-15*, *blaTEM-1B*, *cmlA1*, *mdf(A)*, *sul3*, and *tet(A)*. These findings on the prevalence of ARGs in the APEC MTR_BAU02 genome suggest that the commensal *E. coli* population in poultry in countries with a high and unregulated use of antibiotics constitutes a hotspot for selection for carriage of transferrable resistance. Epidemiological source tracking revealed that the APEC MTR_BAU02 strain has a close evolutionary relationship with many strains of *E. coli* originating from a wide spectrum of clinical samples, including intestinal and extra-

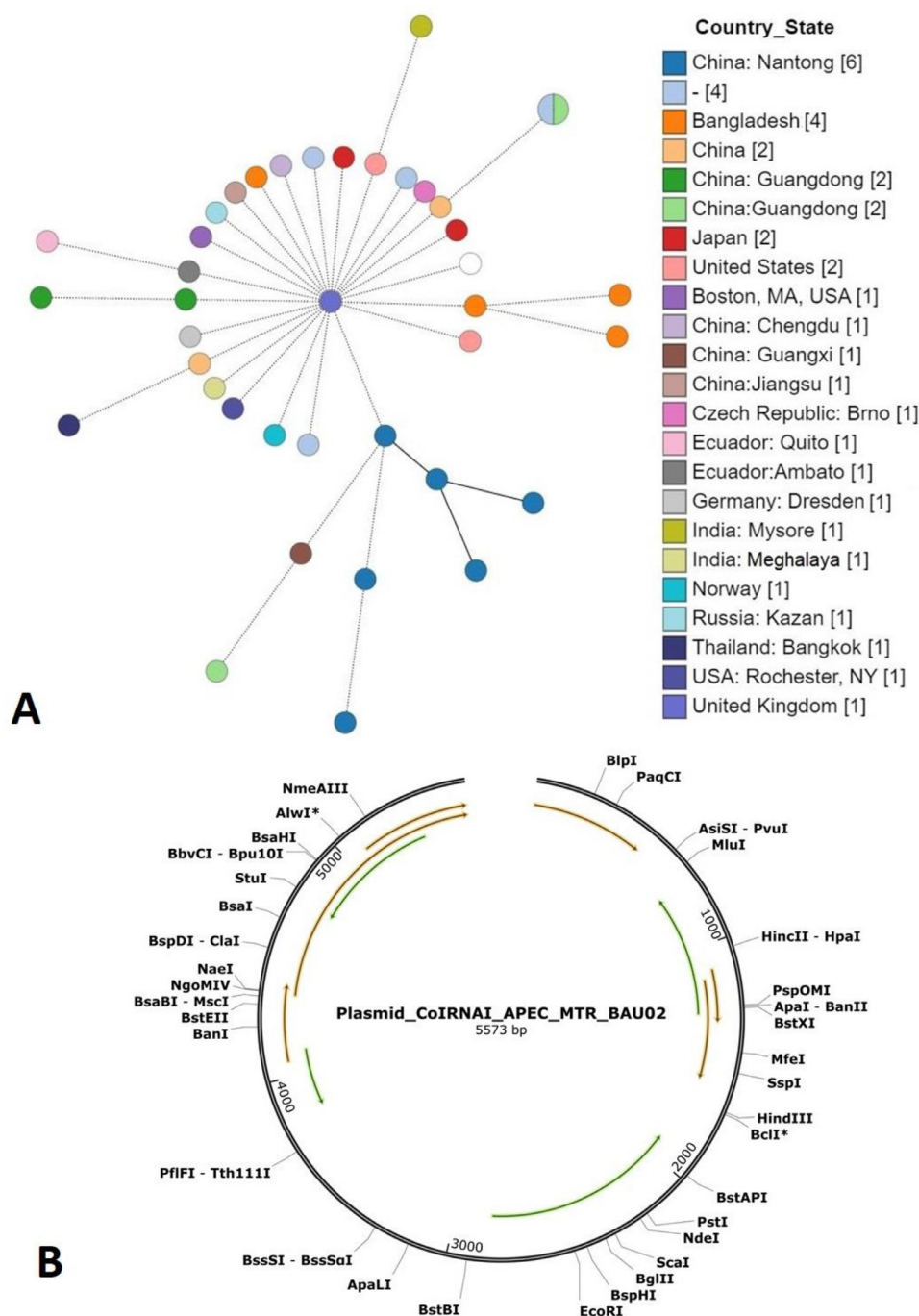


Fig. 2. (A) Phylogenetic relationship between the APEC MTR_BAU02 strain and all *Escherichia coli* strains belonging to ST1196 currently available in the National Center for Biotechnology Information (NCBI) GenBank database through core genome multilocus sequence typing (cgMLST) analysis. (B) Genetic surroundings of plasmid CoIRNAI of the APEC MTR_BAU02 strain. The arrows denote the open-reading frames (ORFs) and restriction sites for different enzymes with coding sequences (CDSs).

intestinal infections (i.e., diarrhea and colibacillosis), urinary tract infections, and community- and hospital-acquired bacteremia in both humans and animals (chickens, cattle, pigs, and dogs). Therefore, APEC strains may represent a zoonotic threat by causing disease in human or animal hosts via horizontal gene transfer of plasmid-linked ARGs and/or VFGs to commensal strains. The CoIRNAI_APEC_MTR_BAU02 plasmid was 5573 bp in size with an average G + C content of 42.4% and 91.15% identity. The plasmid contained 10 predicted open-reading frames (ORFs) and 36 restriction sites for different enzymes with coding sequences (CDSs; Fig. 2B). Basic Local Alignment Search Tool (BLAST) comparison showed

that CoIRNAI_APEC_MTR_BAU02 displayed 100% query coverage and 99.90% identity with virulence plasmid pO26-S3 (the virulence plasmid of Shiga toxin-producing *E. coli*) in *E. coli* strain H30 serovar O26:H11, isolated in August 2008 from a human in the United States.

To interpret the evolution of the APEC MTR_BAU02 strain, a selection of 34 *E. coli* complete genomes downloaded from NCBI was used to map a phylogenetic tree using the maximum-likelihood method. All complete genomes of *E. coli* except our APEC MTR_BAU02 genome were named as NCBI unique IDs. Results showed that the *E. coli* genomes were separated into three major

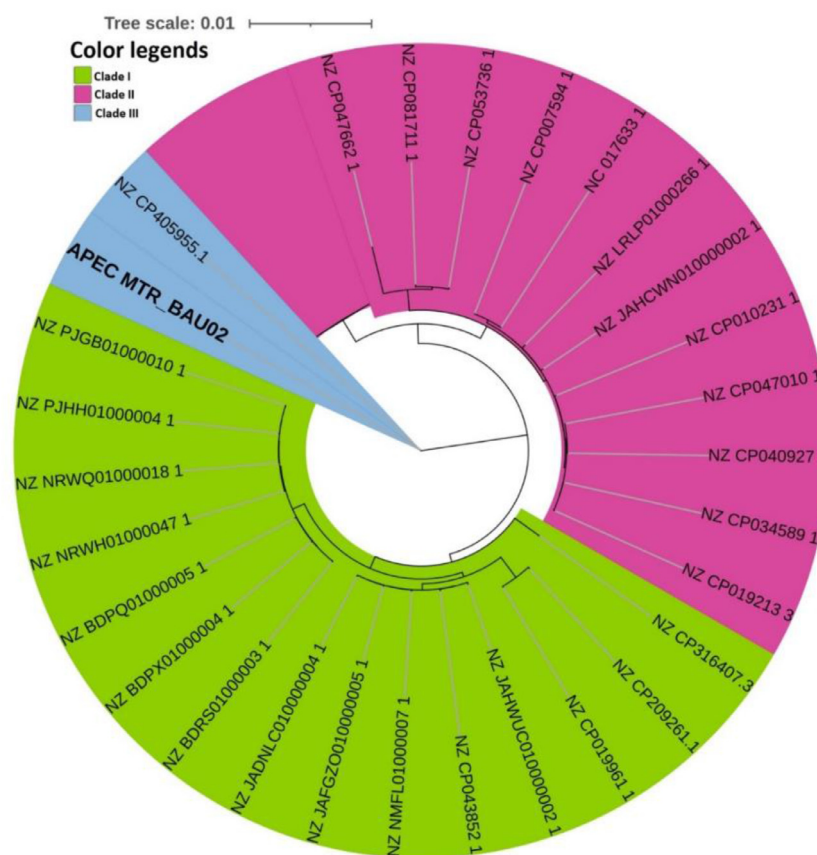


Fig. 3. The evolutionary relationships between APEC MTR_BAU02 and other *Escherichia coli* strains. Thirty-four strains of *E. coli* with complete genomes from NCBI were used for phylogenetic analysis. The mid-point rooted tree was constructed using the NCBI Tree Viewer (<https://www.ncbi.nlm.nih.gov/tools/treeviewer/>) and visualized with iTOL (Interactive Tree Of Life). The evolutionary relationship was inferred using the maximum-likelihood method. Different colors are assigned according to the close evolutionary relatedness (clade) of the genomes. The scale is in the unit of the number of substitutions per site.

clades (I, II, and III). Clades I and II were further subdivided into four subclades. The APEC MTR_BAU02 genome showed the closest evolutionary relationship with APEC O1 (Fig. 3), indicating that APEC MTR_BAU02 may have evolved from avian samples of colibacillosis.

Using VFDB database, 92 VFGs were detected in the APEC MTR_BAU02 genome. Among the identified VFGs, genes coding for iron uptake (25.58%), type II secretion system (20.47%), invasion (16.41%), adherence (12.83%), motility (10.33%), fimbrial activity (7.45%), serum resistance (2.17%), autotransporter (1.80%), and quorum sensing system (1.80%) were the most predominant virulence classes in the APEC MTR_BAU02 genome. The rest of the VFGs had relatively lower (<1.0%) abundances (Fig. 4). The BacWGSTdb 2.0 server detected 49 VFGs, including *E. coli* common pilus structural subunits (*yagZ/ecpA*, *yagX/ecpC*, and *yagY/ecpB*), fimbria-like protein precursor (*fimI*), type 1 fimbriae regulatory proteins (*fimB* and *fimE*), polymerized tip adhesin of ECP (epoxy chloropropane) fibers (*yagW/ecpD*), outer membrane usher protein precursor (*fimD*), and chaperone protein precursor (*fimC*) with >97.0% identity in the genome of APEC MTR_BAU02.

The presumptive SEED functions of the 70 putative genes were predicted through comparison with the nonredundant protein sequences in NCBI database. The physical map of these genes detected in the APEC MTR_BAU02 genome is illustrated in Fig. 5A. Fifty-one of the 70 predicted genes (72.86%) showed high sequence homology to known functional genes with identified SEED functions, ten genes (14.29%) denoted for extracytoplasmic sensor or ligand binding domain (LBD), having po-

tential roles in signal transduction, eight genes (11.43%) encoded for conserved hypothetical proteins with unknown functions, and one gene (1.43%) coded for putative autotransporter (Fig. 5A). The detected functional pathways were distributed under 380 SEED subsystems, which were mostly represented by the genes coding for carbohydrate metabolism (CHO; 17.34%); amino acids and derivatives (14.20%); protein metabolism (10.64%); co-factors, vitamins, prosthetic groups, and pigments (7.49%); respiration (4.72%); membrane transport (4.49%); stress response (4.47%); motility and chemotaxis (4.46%); and virulence, disease, and defense (2.22%). The rest of the SEED subsystems had relatively lower abundances (<2.0%; Fig. 5B). Investigating deeper into SEED subsystem distribution, we found that genes coding for monosaccharide (23.2%) and central CHO (23.18%) metabolism; protein biosynthesis (68.26%); and lysine, threonine, methionine, and cysteine metabolism (30.46%) were top abundant metabolic functional categories in APEC MTR_BAU02. Moreover, flagellar motility (100%); oxidative stress (38.14%); type II protein secretion system (23.71%); chaperon pathway (16.50%); electron donating reactions (55.88%); folate and pterins metabolism (29.63%); and fatty acids, lipids, and isoprenoids (54.54%) were the predominating metabolic features in APEC MTR_BAU02 (Fig. 5B).

Avian colibacillosis is an acute and globally occurring infectious disease of domestic and wild birds caused by different strains of pathogenic *E. coli*, known as APEC strains, and it is associated with considerable economic loss mainly due to high morbidity and mortality rates of the disease [1,4,11]. APEC strains are also im-

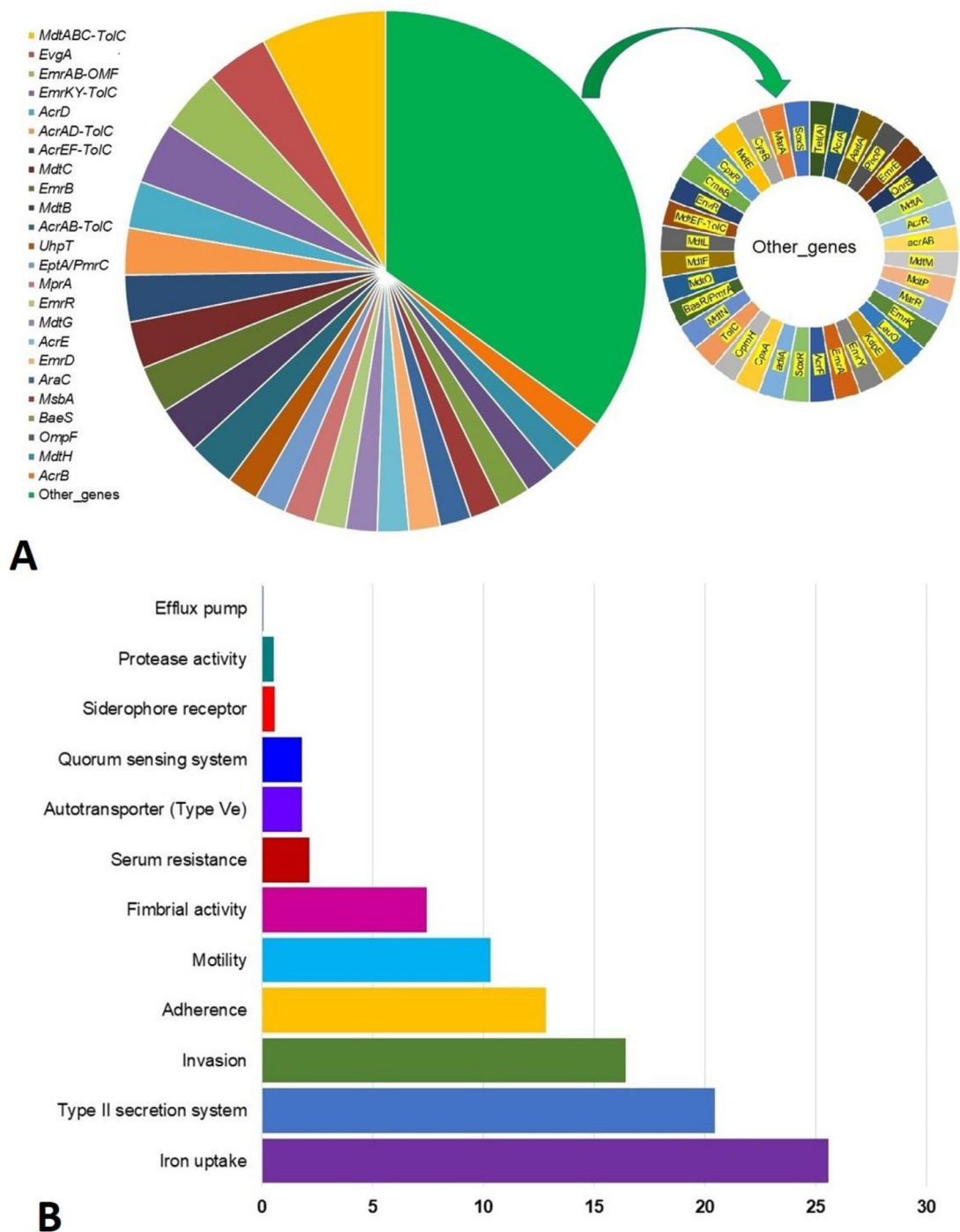


Fig. 4. An overview of the antimicrobial resistance genes (ARGs) and virulence factor genes (VFGs) assigned to the genome of APEC MTR_BAU02. (A) The complete genome sequence of the APEC MTR_BAU02 strain was annotated using the Comprehensive Antibiotic Resistance Database (CARD). The Venn diagram demonstrates the count of each antimicrobial resistance (AMR) feature and the coverage. (B) The complete genome sequence of the APEC MTR_BAU02 strain was annotated using the VirulenceFinder database. The bar plots demonstrate the count of each VFG feature and the coverage.

portant from a public health perspective. To control colibacillosis, multiple antimicrobials have been used indiscriminately, especially in middle- and low-income countries including Bangladesh, contributing to the development and spread of AMR and/or ARGs [4,10]. The subsequent selection of MDR strains has generated serious challenges in terms of public health. Thus, investigations of APEC strains regarding their phylogenetic evolution, metabolic functional potentials, and VFG and ARG profiles may help curtail

their hazardous effects. Colibacillosis in poultry farms might be prevented and/or controlled by the rational therapeutic use of antimicrobials. However, evolution of MDR APEC strains along with the transmission of resistance genes has created challenges in reducing the risk of APEC infections [1].

In summary, this is the first WGS-based study on a MDR APEC strain isolated from layer chicken in Bangladesh. In this study, we report the genomic characteristics of a MDR *E. coli* strain carrying

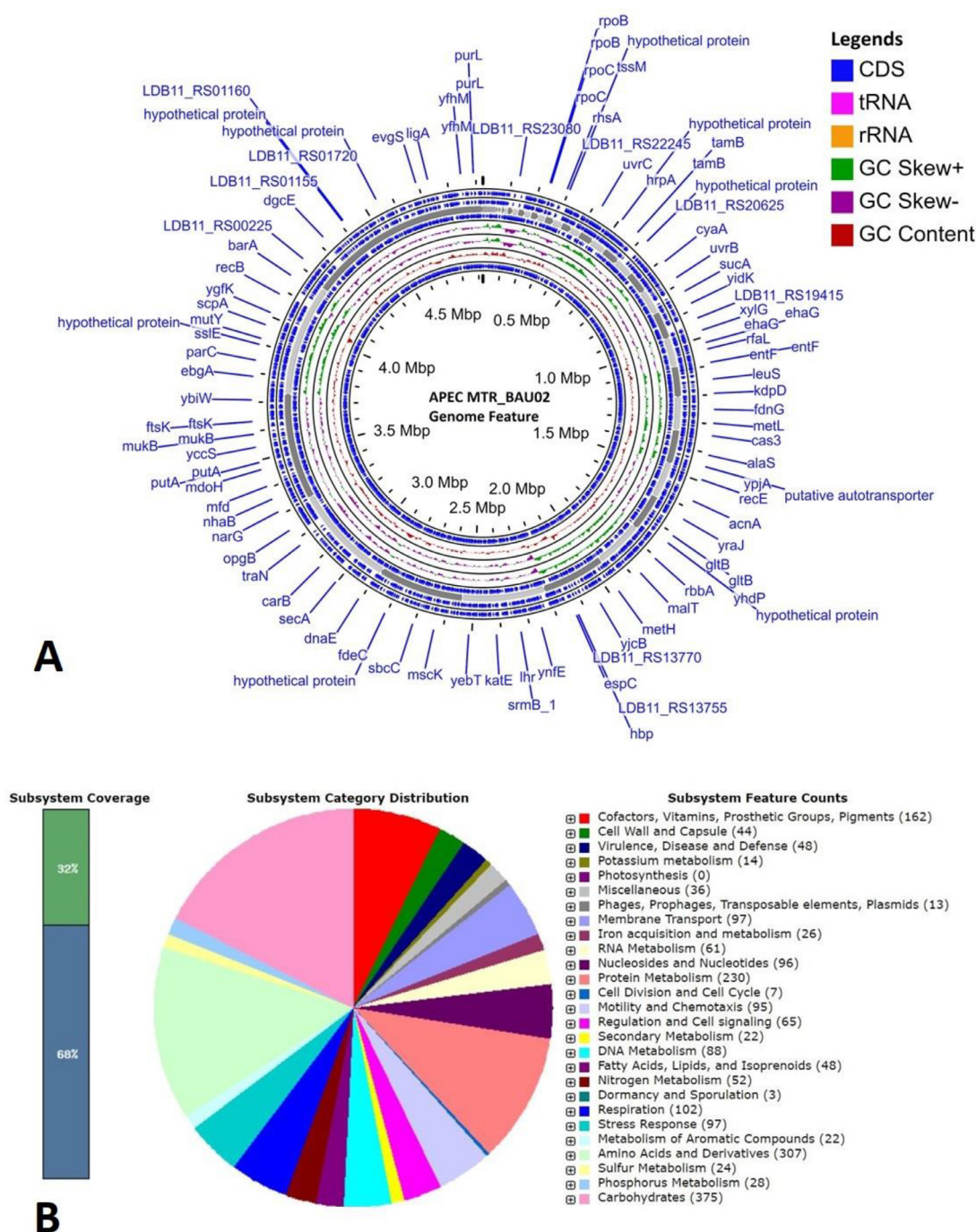


Fig. 5. (A) BRIG- and CGView-derived schematic map of the circular genome of APEC MTR_BAU02. The open-reading frames (ORFs) with predicted annotations are indicated with blue arrows in the innermost ring, which shows the genome location. The innermost ring is followed by rings representing G + C content (dark red), and G + C skew + (green) and - (purple). The two most external rings show identified coding sequences (CDSs) on both strands represented by blue arrows, and results of the genome annotation process. The putative function of each CDS is labelled at the outermost ring, and the organization of the SEED functional modules of 70 of the CDSs are represented at the innermost ring. (B) Subsystem categorical distribution in the APEC MTR_BAU02 genome. The whole genome sequence of the APEC strain MTR_BAU02 was annotated using the Rapid Annotation System Technology (RAST) server. The pie chart demonstrates the count of each subsystem feature and the subsystem coverage.

different ARGs, VFGs, and genomic functional potentials which may help to elucidate the dissemination mechanisms of AMR genes amongst bacteria, animals, and humans. These results call for immediate action from policy makers to stop imprudent use, or actively regulate rational use, of antibiotics, as well as other important antimicrobials in poultry production in Bangladesh. These data can facilitate an unravelling of the genomic features, antimicrobial resistance mechanisms, and epidemiological characteristics of this bacterial pathogen.

Nucleotide sequence accession number

The WGS data of the APEC MTR_BAU02 strain has been deposited in GenBank under accession number JAIWJB000000000, and the assembly reports of the genome are also available in GenBank. The version described in this paper is version JAIWJB000000000.1. The Illumina shotgun reads are available in the National Center for Biotechnology Information (NCBI) Sequence

Read Archive (SRA) under accession number SRR16193628, and under BioProject accession number PRJNA766218.

Funding

This research project was funded by the Ministry of Education, Government of the People's Republic of Bangladesh (Project No. LS2018686).

Conflicts of interest

None declared

Ethical approval

Not required

Acknowledgments

We are deeply grateful to the Ministry of Education, Government of Bangladesh for funding the research project (Project No. LS2018686). In addition, we are grateful to the poultry farm owners for giving access to the samples. The authors would also like to thank the Department of Microbiology and Hygiene, Faculty of Veterinary Science Bangladesh Agricultural University, Mymensingh for their support during the research.

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